INCH-POUND
MIL-DTL-7793/15A
19 October 2012
SUPERSEDING
MIL-M-7793/15
15 December 1988

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL,
115 VAC, 50 TO 2400 Hz, SQUARE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Reactivated after 19 October 2012 and may be
used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

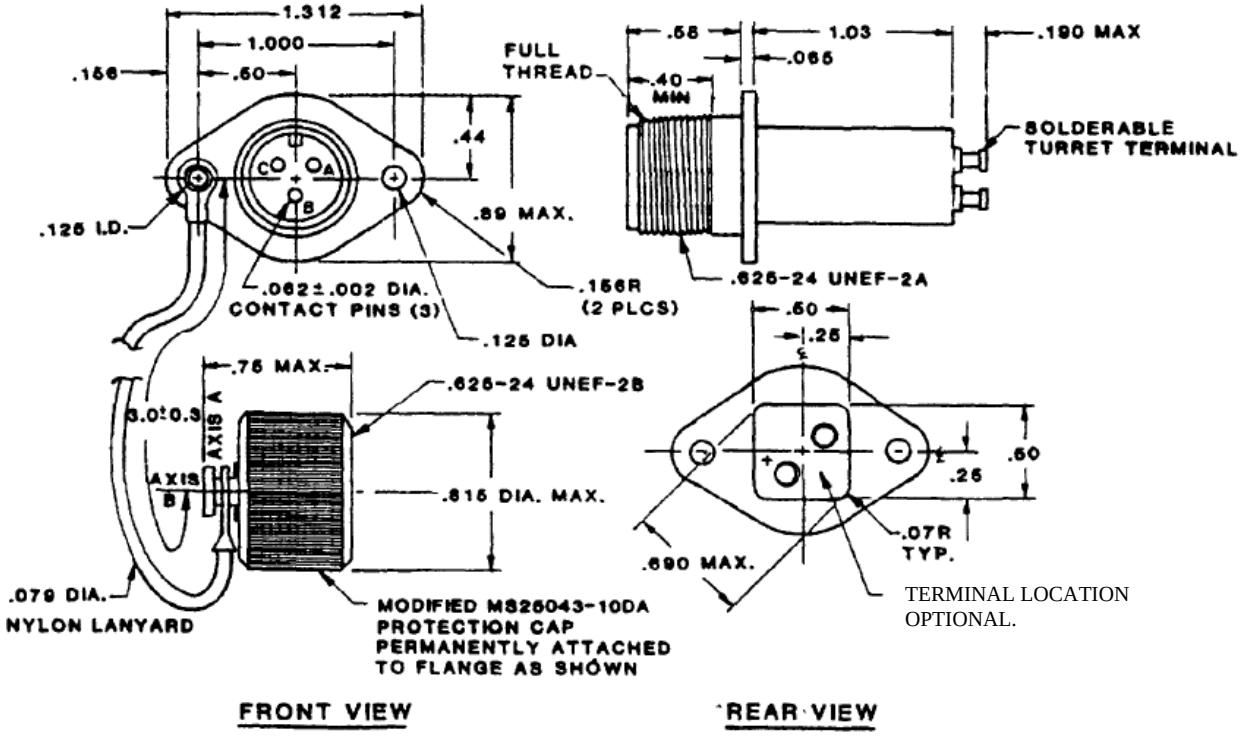


FIGURE 1. M7793/15-1.

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REQUIREMENTS:

1. Dimensions. Dimensions and configuration shall be in accordance with figure 1. All dimensions are in inches. Unless otherwise specified, tolerances are ± .02 for two-place decimals and ±0.015 for three-place decimals.

2. Mounting flange. The mounting flange, threaded extension and threaded protection cap shall be constructed of aluminum alloy with an olive drab chromate coating over cadmium plating.

3. Protection cap. The protection cap shall be in accordance with MS25043-10DA per SAE-AS25043 except for the finish and the means of attachment. The finish shall be in accordance with Requirement 2 and the metallic chain shall be replaced with a nylon lanyard.

4. Contacts. The contacts shall meet all the requirements of silver plated solder contacts from SAE-AS50151.

PERFORMANCE REQUIREMENTS:

1. Dielectric strength. The meter shall be capable of withstanding 600 VRMS, 60 Hz (room) and 350 VRMS, 60 Hz (altitude).

2. Life accuracy. Life accuracy shall be ±0.1% from -65° to +125° C and 75 to 150 VAC at 50 to 2400 Hz.

3. Power consumption. Power consumption at 115 VAC, 60 or 400 Hz, shall not exceed 50 milliwatts.

4. Operating temperature range. The temperature range shall be from -65° C to +125° C.

5. Part number. The part number is M7793/15-1.

6. Total readout. Total readout of the meters shall be 99999.99 hours.

7. Meter output. The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10%. A logical “zero” shall be between 0.0 and +0.2 volts. A logical “one” shall be between +3.3. and +6.6 volts. The output pin assignments are shown on figure 2 and the output data format is shown on figure 3.

8. Transient protection. No temporary or permanent degradation shall be produced in the meter if the input voltage should increase to 180 VRMS at 50 to 2400 Hz for a period of 150 milliseconds maximum.

9. Durability of cap and threads. No mechanical or electrical degradation after 100 cycles in accordance with SAE-AS50151.

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10. Voltage range. Operating voltage shall be from 75 to 150 VAC.

11. Durability of connectors and contacts. No mechanical or electrical degradation after 100 cycles in accordance with SAE-AS50151. Operational accuracy shall be checked at the end of the test.

12. Weight. Weight shall be not greater than 1.0 ounce with the attached protection cap.

13. Tensile strength of lanyard. The lanyard shall withstand a tensile load of 25 pounds when applied to Axis “A” and Axis “B” as shown on figure 1.

14. Qualification tests. Qualification tests will be conducted in accordance with MIL-DTL-7793, except that the dielectric strength, voltage, temperature and accuracy shall be as specified herein. Salt spray, moisture resistance, vibration, shock, electromagnetic compatibility and low temperature storage shall be conducted with the protective cap in place. Hermetic seal test is not applicable.

NOTES:

1. Elapsed time is read from the M7793/15-1 meter by mating the M7793/15-1 meter’s connector with the M7793/12-1 reader’s connector and operating the reader.
2. Digital readout and meter face plate per MIL-DTL-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.
3. The meter is design to have an electrical ground and a continuous EMI/RFI shield with the threaded protection cap in place.
4. AC operation may be from either sinusoidal or square wave sources.

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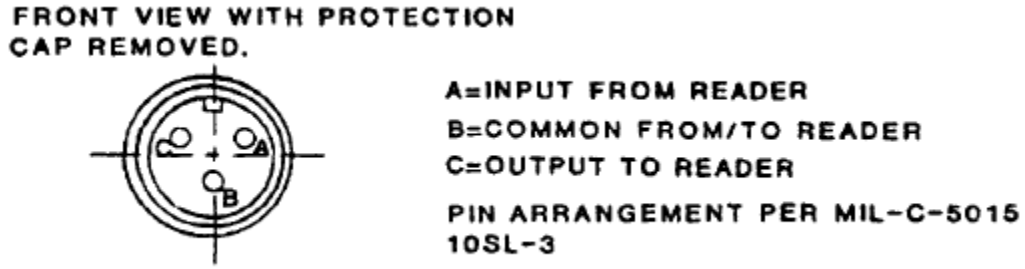


FIGURE 2. Readout pin assignment.

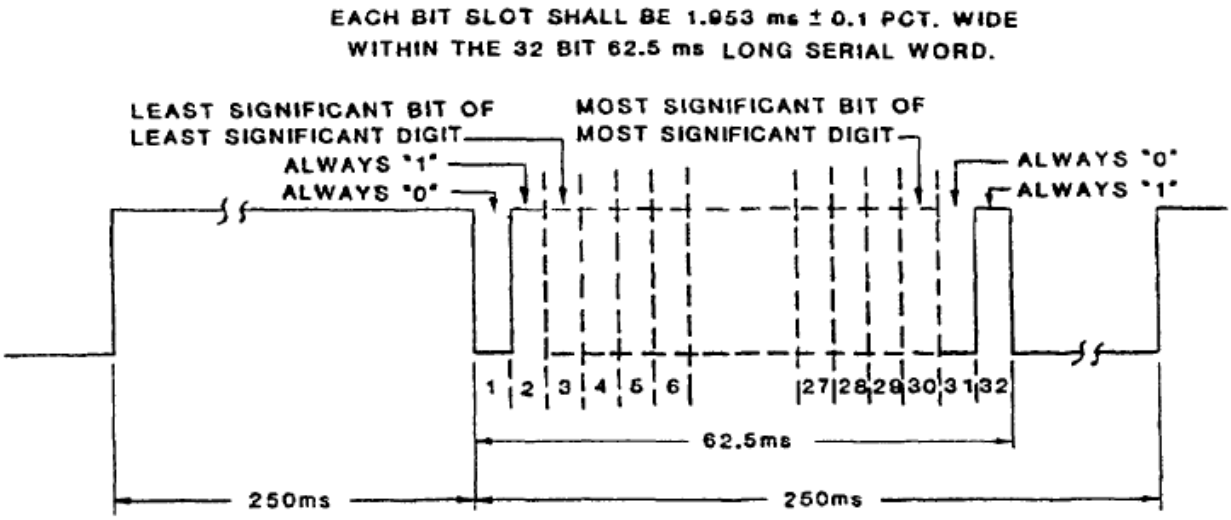


FIGURE 3. Output data format.

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8. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-7793, this specification sheet references SAE-AS50151 and SAE-AS25043.

Custodians:
Army - MI
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS
(Project 6645-2012-017)

Review activities:
Navy – OS, SH
DLA – GS1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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